ISC 211, Introduction to Bioinformatics

Assignments (Individual Assignments)

Deadline: 15-10-2026 (11:59 PM)

You are assigned to implement the biological problem $Topic_i$, using the machine learning algorithm M_j aim to understand the application of machine learning techniques to biological data analysis. The allocation should be carried out based on your roll number using the following formulas:

$$Topic_i = Topic_{((R-1) mod\ 11)+1}$$

$$M_j = \left\{ M_{((R-1) mod\ 13)+1} \ if\ Topic_i \leq 10\ else\ one\ from\ \{M_{11}, M_{12}, M_{13}\} \right.$$

$where$:

$$R = \ Last\ \ three\ digit\ of\ your\ roll\ number$$

$$Topic_i = \ Assigned\ Asssignmnet\ topic\ from\ the\ list\ of\ 11\ topics$$

$$M_j = \ Assigned\ Machine\ Learning\ algorithm\ from\ the\ list\ of\ 13\ algorithms$$

Topics

1. Machine Learning-Based Cancer Classification Using Gene Expression Data (https://www.ijml.org/vol14/IJML-V14N2-1157.pdf)
2. DNA Sequence Classification Using Machine Learning (https://ieeexplore.ieee.org/abstract/document/11042647)
3. Protein Family Classification Using Machine Learning (https://link.springer.com/article/10.1007/s10930-024-10181-5)
4. Promoter Region Prediction Using Machine Learning (https://ieeexplore.ieee.org/abstract/document/10601371)
5. Disease Prediction Using Gene Expression Analysis (https://ieeexplore.ieee.org/abstract/document/9693112)
6. Drug Response Prediction Using Machine Learning(https://link.springer.com/chapter/10.1007/978-981-16-9221-5_20)
7. Biomarker Discovery Using Machine Learning (https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1010357)
8. Protein Function Prediction using Machine Learning (https://pubmed.ncbi.nlm.nih.gov/31603244/)
9. Protein Structure Classification Using Machine Learning (https://www.sciencedirect.com/science/article/pii/S0959440X2300101X)

10. Mutation Impact Prediction Using Machine Learning (https://pmc.ncbi.nlm.nih.gov/articles/PMC10239318/)
11. Gene Expression Clustering and Analysis (https://ceur-ws.org/Vol-3372/paper02.pdf)

Algorithms:

Algorithm ID	Algorithm
M_1	Logistic Regression
M_2	Support Vector Machine (SVM)
M_3	Random Forest
M_4	Decision Tree
M_5	k-Nearest Neighbors (k-NN)
M_6	Naïve Bayes
M_7	XGBoost
M_8	Gradient Boosting
M_9	Artificial Neural Network (ANN)
M_{10}	Convolutional Neural Network (CNN)
M_{11}	K-Means Clustering
M_{12}	Hierarchical Clustering
M_{13}	Principal Component Analysis (PCA)

Task

- Read and understand the assigned research paper.
- Identify the biological problem addressed
- Identify the dataset used in the paper or select a suitable public dataset.
- Prepare the dataset for machine learning implementation.
- Implement the assigned machine learning algorithm.
- Train the model using the biological dataset.
- Test the model using unseen data and evaluate their model using appropriate metrics.
- Analyse the results
- Prepare the report

Submission Procedure:

- Upload the zip file containing the source code, dataset and the report (Max 5 pages including the graphs, tables etc) on the LMS